

$$V_{upper} = -22 \text{ V}$$

20

C

$$\text{For X, } R_X = \frac{\rho L}{A} = \frac{\rho l}{l^2}$$

$$\text{For Y, } R_Y = \frac{\rho L}{A} = \frac{2\rho l}{2l^2} = R_X$$

21

A

$$I = nAve$$